



Accessing a Home Laptop Remotely From Android

Consider this: You're on vacation in the Maldives, when your office calls, with an urgent request for some files and information. These happen to be on your laptop, which is at home! If you've had a holiday ruined like this, here are some simple steps that help you connect to your remote laptop securely, if you have left it on and connected to the Internet—or if someone at home can switch it on for you.

For clarity, we have used a scenario where you have a smartphone running Google Android, and will use that to access your laptop, which is running Ubuntu Lucid. The laptop is connected to the Internet via a broadband connection, and the phone has some wireless access (Wi-Fi, 3G, GPRS, etc) to the Internet. The basic idea in this article can, however, be extended to different operating systems and client devices.

So let's go through the steps in the procedure to set up remote access in this scenario.

Get a DNS name for your home network

1. You can have someone at home determine the external IP address (assigned by your ISP to your broadband modem/router) using a website like <http://www.whatismyip.com/> [1]. However, handling IP addresses is somewhat more difficult, compared to using host-names—say, *laila.dyn dns-server.com*. Also, some ISPs often automatically reset your dynamic IP address, even without you rebooting your router. Tracking the changing IP in this case becomes a challenge. Is there a simple workaround to this problem?

Yes! With dynamic DNS update services, such as DynDNS [2], you can create a free account, and assign a unique host-name to your external IP address. This way, you don't need to juggle IP addresses. To update the free DNS service with the latest-assigned IP address, a scheduled job should run on a machine in your home network, at intervals you specify (an hour or so, perhaps).

2. In our scenario, we created and placed a *dyn dns* file in the

/etc/cron.hourly/ folder. The script, along with detailed instructions on setting up and working with DynDNS, is available at https://www.dyn dns.com/about/home_solutions.html [3].

Set up the router for external access

Each machine in your home network will usually be assigned a unique, private IP address that is not directly accessible from the Internet. To allow incoming requests from the Internet to reach your laptop, you need to do some configuration on the router, which is the gateway between your private network and the Internet. This is done in the administrative interface for the router. Some routers have it under the Network Address Translation (NAT) tables for the router, some in other pages. You could look up the documentation for your router to learn how to set up a *port forwarding* rule; before you can set it up, you need to have the following configuration done, so you know the target internal IP address and port number for the rule.

Static IP address for the laptop

Most home networks use IP addresses that are automatically assigned by the DHCP service on the broadband router. In this case, depending on the order in which systems are started up, the IP address assigned to your laptop could change, which means your port-forwarding rule will direct incoming connections to the wrong computer. Thus, it is important to give your laptop a static IP address. In our scenario, we will set a static IP address in Ubuntu using simple command-